Prabhavit Gupta

Research Engineer | LLM Pre-Training, PyTorch, HuggingFace & Distributed Inference

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Executive Summary

- **LLM Pre-Training & Synthetic Dataset Engineering:** AI Engineer and applied researcher with 3 years commercial software experience specializing in transformer architectures, tokenizer optimization, and large-scale dataset curation for coding models. Deep practical focus on scalable pre-training pipelines and automated evaluation harnesses.
- **Deep Learning Infrastructure & High-Throughput Inference:** Hands-on expertise with PyTorch, HuggingFace Transformers, vLLM, and distributed GPU acceleration. Proven record designing continuous model verification gates, quantization benchmarks, and low-latency inference runtimes for developer tooling.
- **Autonomous Tooling & Developer Assistance Systems:** Experienced taking foundation models from training clusters to production microservices. Strong foundation in statistical learning, loss landscape tracking, and multi-agent coordination architectures with extreme engineering ownership.

Technical Skills & Core Competencies

- **Deep Learning & LLMs:** PyTorch, HuggingFace (Transformers, Datasets, Accelerate), DeepSpeed, vLLM, TensorRT-LLM, LoRA/QLoRA Fine-Tuning, KV-Cache Optimization.
- **Data Curation & Tokenization:** High-Throughput Code Ingestion Pipelines, Deduplication, AST-Based Code Filtering, Synthetic Data Generation, Custom BPE Tokenizer Training.
- **Evaluation & Observability:** LLM Benchmarking (HumanEval, MBPP, RAGAS), Loss Tracking & Drift Detection, Weights & Biases, Langfuse Distributed Tracing.
- **Core Engineering:** Python (AsyncIO, CPython Profiling, PyTest), Docker, Kubernetes basics, Git, GitHub Actions CI/CD, TeamCity familiarity.
- **Research Methodology:** First-Principles Reasoning, Empirical Ablation Studies, Reproducibility, Clean Modular Code.

Professional Experience

AI Research Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and executed high-scale data ingestion and synthesis pipelines preparing multi-gigabyte code and reasoning datasets for domain-specific language model adaptation.
- Built automated regression and capability evaluation harnesses benchmarking code generation accuracy, syntactic correctness, and execution consistency across model iterations.
- Optimized production inference architectures using vLLM and PagedAttention, achieving a 45% increase in serving throughput and sub-100ms first-token latency.
- Implemented distributed trace observability using Langfuse to track latency, token economics, and error modes across real-world developer assistant workflows.
- Collaborated with European engineering teams to translate complex architectural specifications into scalable, production-grade PyTorch and Python microservices.
- Authored modular evaluation harnesses integrated into automated CI/CD workflows, ensuring prompt and weights updates do not introduce behavioral regressions.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Engineered high-throughput asynchronous data processing backends in Python (FastAPI) and TypeScript for structured extraction from complex code and document corpuses.
- Implemented semantic vector retrieval pipelines over PostgreSQL (pgvector) and Redis, accelerating context lookup speeds under concurrent load.
- Established automated unit, integration, and performance benchmarking test suites with PyTest, enforcing rigorous release standards.
- Worked across the engineering lifecycle from algorithmic experimentation to Docker containerization and cloud deployment.

Key Engineering Projects

- **High-Throughput Code Dataset Curation & AST Tokenizer (Python + PyTorch + Tree-Sitter):** Developed an automated pipeline parsing and filtering open-source code repositories using abstract syntax trees to eliminate duplicate and low-quality code for pre-training corpora.

- **Continuous Model Evaluation Harness for Coding Benchmarks (PyTorch + vLLM + PyTest):** Implemented an isolated sandbox evaluating generated code solutions against test suites (HumanEval/MBPP format) with automated latency and pass@k metric tracking.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in mathematical logic, statistical decision theory, computational semantics, and formal reasoning.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), German (Basic/Elementary).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union (including Germany, Spain, Netherlands, Czech Republic) without visa sponsorship. Immediate availability.